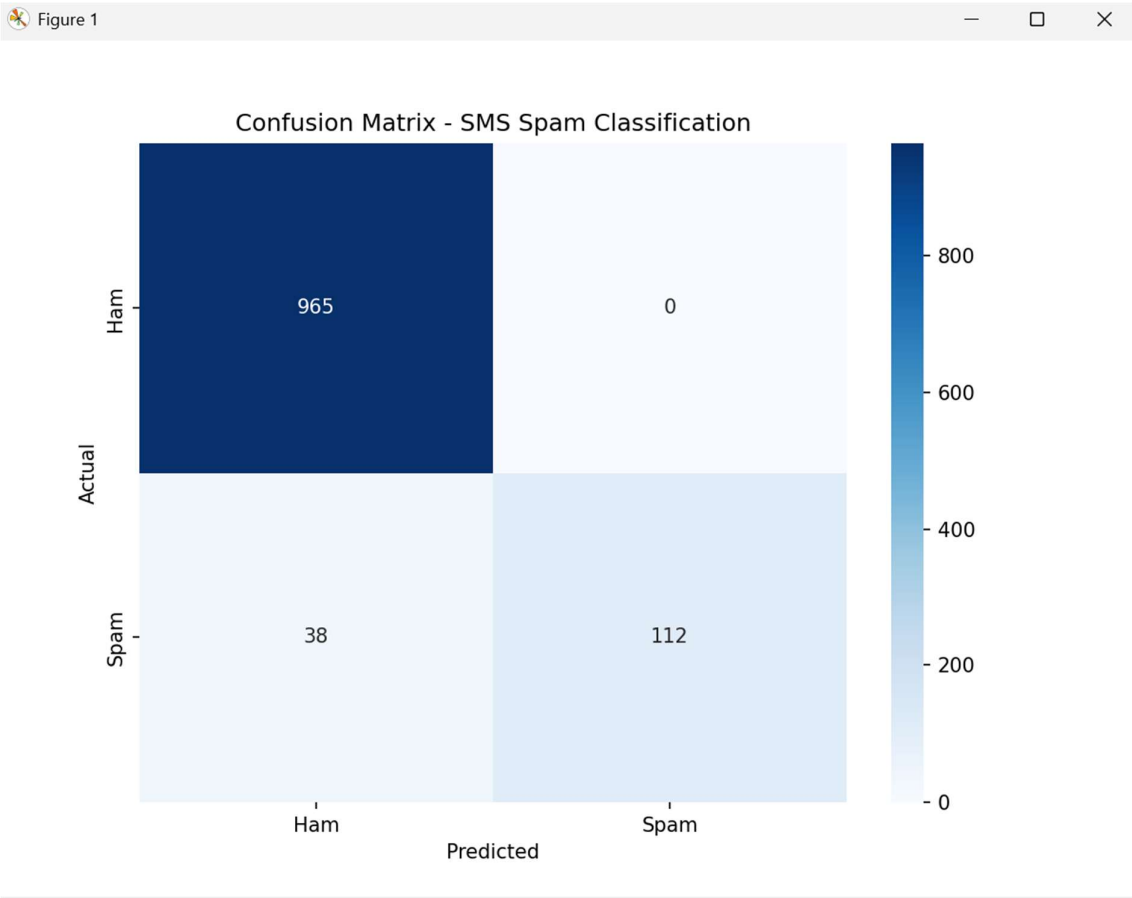
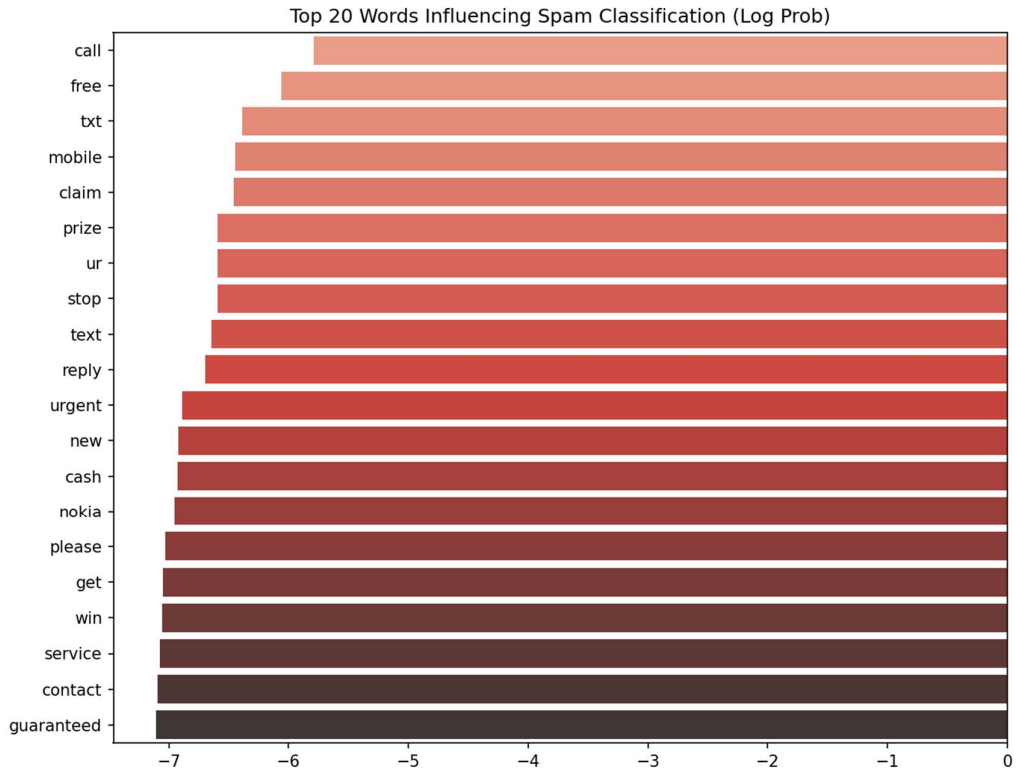
ML – ASSIGNMENT 5– OUTPUT

SCENARIO 1




Top 20 Words Influencing Spam Classification (Log Prob)
call
free
txt
mobile
claim
prize
ur
stop
text
reply
urgent
new
cash
nokia
please
get
win
service
contact
guaranteed
−7
−6
−5
−4
−3
−2
−1
0

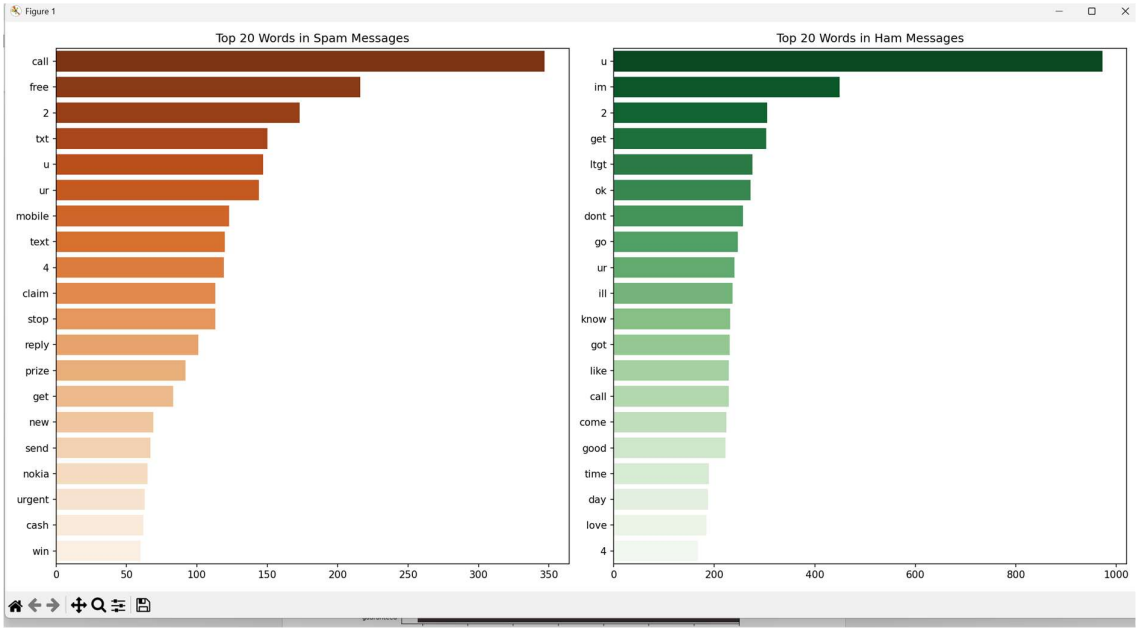


Figure 1
Top 20 Words in Spam Messages
call
free
2
txt
u
ur
mobile
text
4
claim
stop
reply
prize
get
new
send
nokia
urgent
cash
win
0
50
100
150
200
250
300
350
Top 20 Words in Ham Messages
u
im
2
get
ltgt
ok
dont
go
ur
ill
know
got
like
call
come
good
time
day
love
4
0
200
400
600
800
1000

SCENARIO : 2

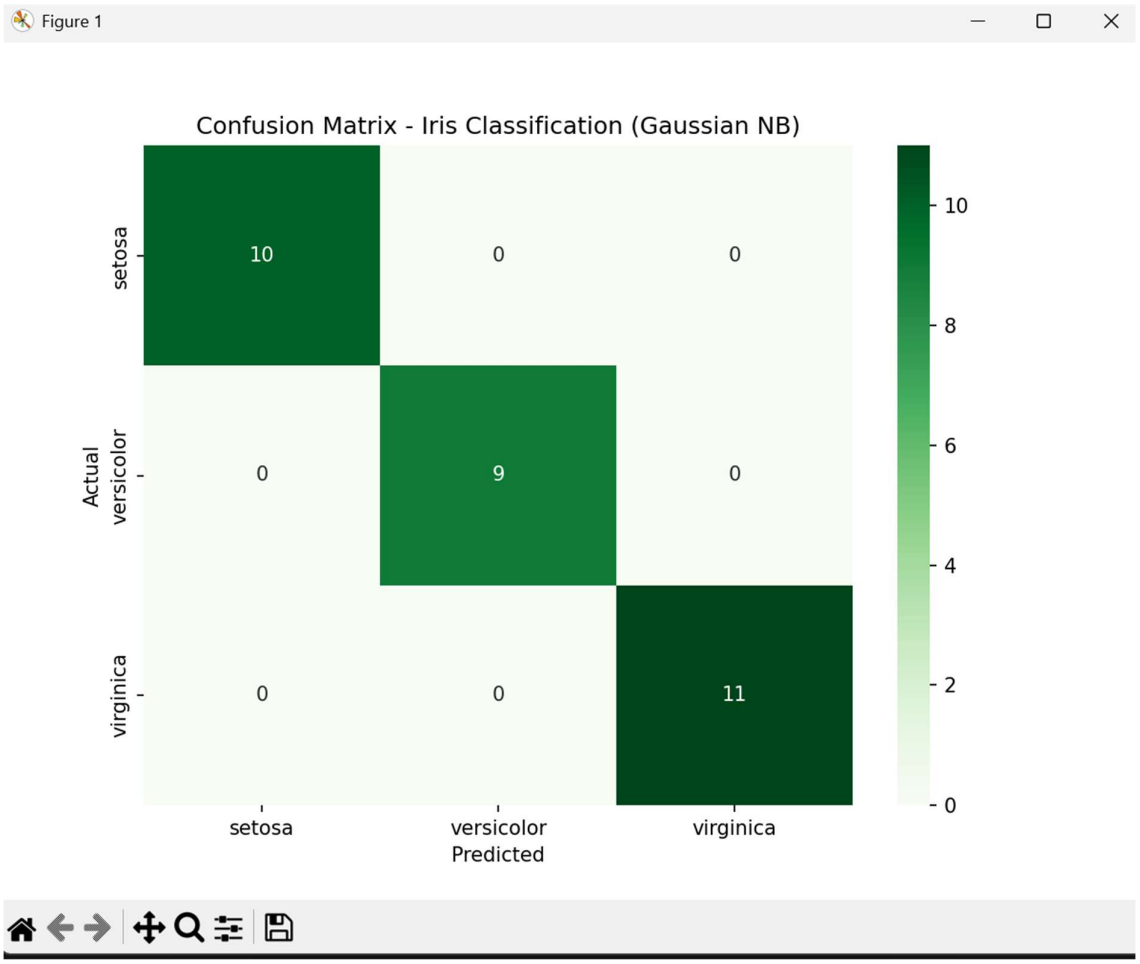
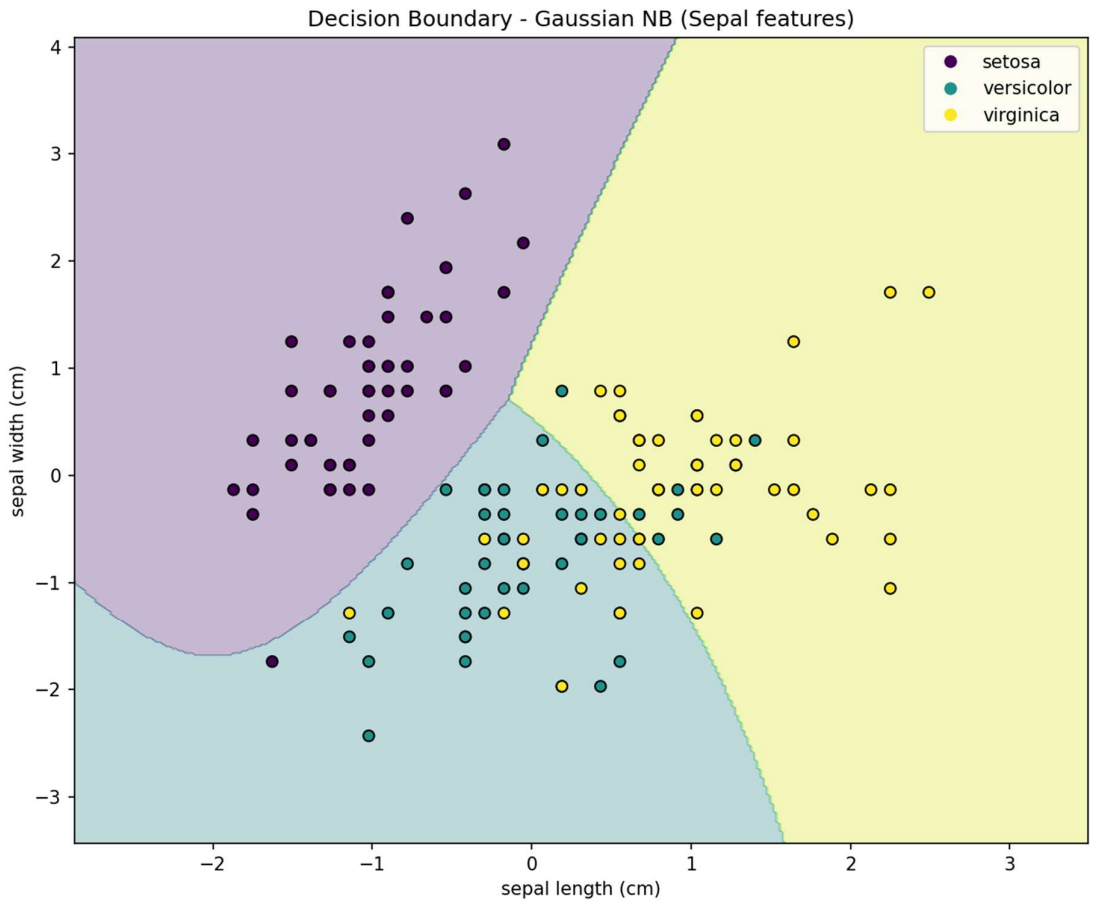
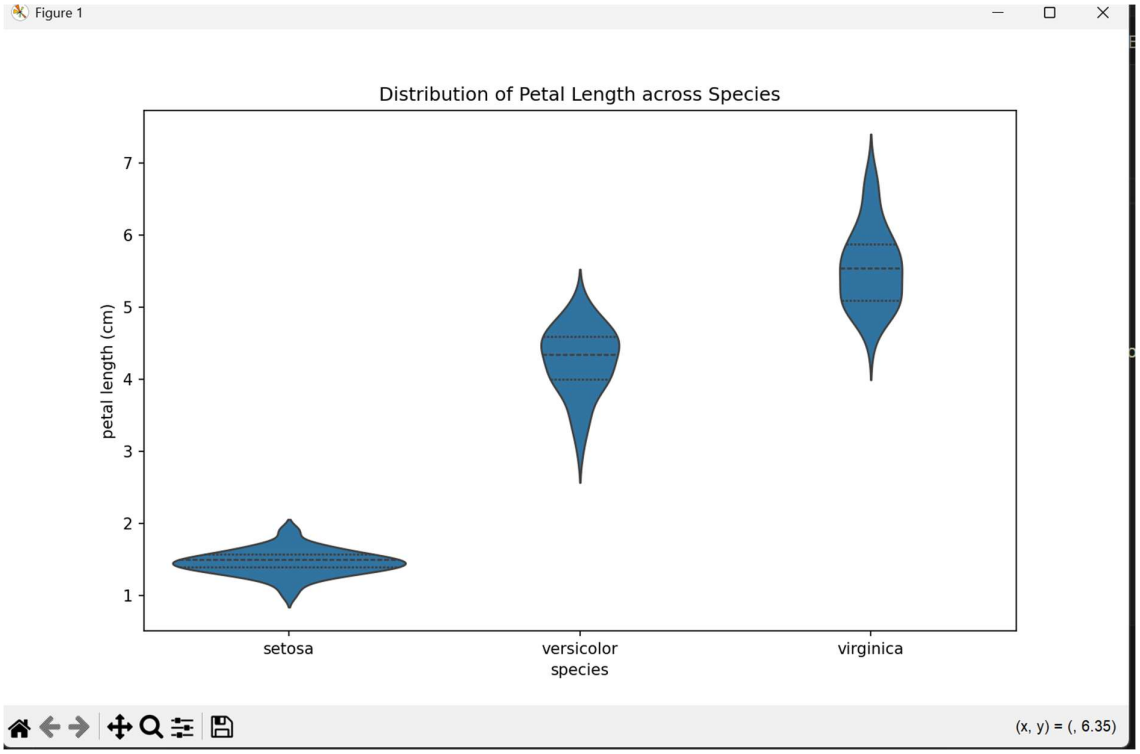


Figure 1
Confusion Matrix - Iris Classification (Gaussian NB)
Actual
setosa
versicolor
virginica
10
0
0
0
9
0
0
0
11
setosa
versicolor
virginica
Predicted
10
8
6
4
2
0


Decision Boundary - Gaussian NB (Sepal features)
setosa
versicolor
virginica
sepal width (cm)
sepal length (cm)
4
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3

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Iris Dataset Inspection (Head):
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  species
0                5.1               3.5                1.4               0.2        0
1                4.9               3.0                1.4               0.2        0
2                4.7               3.2                1.3               0.2        0
3                4.6               3.1                1.5               0.2        0
4                5.0               3.6                1.4               0.2        0

Description:
       sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)     species
count         150.000000        150.000000         150.000000        150.000000  150.000000
mean            5.843333          3.057333           3.758000          1.199333    1.000000
std             0.828066          0.435866           1.765298          0.762238    0.819232
min             4.300000          2.000000           1.000000          0.100000    0.000000
25%             5.100000          2.800000           1.600000          0.300000    0.000000
50%             5.800000          3.000000           4.350000          1.300000    1.000000
75%             6.400000          3.300000           5.100000          1.800000    2.000000
max             7.900000          4.400000           6.900000          2.500000    2.000000

Training Gaussian Naive Bayes...

Gaussian NB Performance:
Accuracy: 1.0000
Precision (Weighted): 1.0000
Recall (Weighted): 1.0000
F1 Score (Weighted): 1.0000

Sample Comparisons (First 10):
       Actual   Predicted
0  versicolor  versicolor
1      setosa      setosa
2   virginica   virginica
3  versicolor  versicolor
4  versicolor  versicolor
5      setosa      setosa
6  versicolor  versicolor
7   virginica   virginica
8  versicolor  versicolor
9  versicolor  versicolor
```

```
3  versicolor  versicolor
4  versicolor  versicolor
5      setosa      setosa
6  versicolor  versicolor
7   virginica   virginica
8  versicolor  versicolor
9  versicolor  versicolor

Class Probabilities for First 5 Test Samples:
          setosa    versicolor     virginica
0   5.973226e-90  9.956358e-01  4.364232e-03
1   1.000000e+00  4.961577e-14  6.549219e-21
2  7.318611e-290  4.929473e-12  1.000000e+00
3   2.818383e-94  9.775936e-01  2.240644e-02
4  1.138764e-105  8.700226e-01  1.299774e-01
```